Unit 9

Partial Differential Equation and Fourier Series

Two-point Boundary Value Problems

Initial Value Problem (IVP)

An initial value (IVP) of second order is a differential equation

$$y'' + p(t)y' + q(t)y = g(t) \quad ... \quad (1)$$

together with the conditions:

$$y(t_0) = y_0, y'(t_0) = y'_0 \quad ... \quad (2)$$

The conditions in (2) are called the initial conditions.

Boundary Value Problems

A boundary value problem (BVP) (Two Point) is a differential equation

$$y'' + p(x)y' + q(x)y = g(x) \quad ... \quad (3)$$

together with the conditions

$$y(\alpha) = y_0, y(\beta) = y_1 \quad ... \quad (4)$$

The conditions (4) are called the boundary conditions.

Homogeneous BVP

The boundary value problem (3) is said to be homogeneous if $g(x) = 0$ for every value of x and if $y_0 = 0$, $y_1 = 0$ otherwise, it is called non-homogeneous BVP.

Examples

$i.$ $y'' + y = 0, y(0) = 0, y'(t) = 1$ (non-homogeneous)

$ii.$ $y'' + 3y = cosx, y(0) = 0, y'(\pi) = 0$ (non-homogeneous)

$iii.$ $y'' + hy = 0, y'(0) = 0, y(s) = 0$ (homogeneous)

$iv.$ $x^2y'' - xy' + \lambda y = 0, y(1) = 0, y(l) = 0,$ (homogeneous)

Solution of Boundary Value Problem

We first find the general solution of the differential equation and then use the boundary conditions to find the value of arbitrary constants. Substituting these values of arbitrary constants is the general solution, we get the required solution of the BVP.

Note

The homogeneous BVP has a trivial solution. The non-homogeneous BVP has a unique solution iff the corresponding homogeneous BVP has only the trivial solution, that is 0 solution. And the non-homogeneous BVP has either no solution or infinitely many solution iff the homogeneous BVP has non-trivial solutions.

Example (Non homogeneous BVP with unique solution)

Solve the BVP

$$y'' + 2y = 0, y(0) = 1, y(\pi) = 0$$

Solution

The given BVP is

$$y'' + 2y = 0 \quad ... \quad (1)$$

together with

$$y(0) = 1, y(\pi) = 0 \quad ... \quad (2)$$

The A.E. of (1) is

$$m^2 + 2 = 0$$

$$\therefore m = \pm\sqrt{2}i = 0 \pm \sqrt{2}i$$

So, the general solution of (1) is

$$y = e^{0x}\left(c_1 cos\sqrt{2}x + c_2 sin\sqrt{2}x\right)$$

$$\Longrightarrow y = c_1 cos\sqrt{2}x + c_2 sin\sqrt{2}x \qquad \dots \qquad (3)$$

Using the boundary condition

$$y(0) = 1,$$

we get,

$$1 = c_1 + 0 \therefore c_1 = 1.$$

Again, using

$$y(\pi) = 0,$$

we get

$$0 = c_1 cos\sqrt{2}\pi + c_2 sin\sqrt{2}\pi$$

$$\Longrightarrow 1. cos\sqrt{2}\pi + c_2 sin\sqrt{2}\pi = 0$$

$$\Longrightarrow c_2 = -\frac{cos\sqrt{2}\pi}{sin\sqrt{2}\pi} = -cot\sqrt{2}\pi = -0.28$$

$$\therefore c_2 = -0.28$$

Hence, the required solution of the BVP is

$$y = cos\sqrt{2}x - 0.28 sin\sqrt{2}x$$

Example

Solve the boundary value problem

$$y" + y = 0, y(0) = 1, y(\pi) = a.$$

Solution

The given boundary value problem is

$$y" + y = 0 \qquad \dots \qquad (1)$$

With boundary conditions

$$y(0) = 1, y(\pi) = a \qquad \dots \qquad (2)$$

The auxiliary equation of (1) is

$$m^2 + 1 = 0$$

$$\Longrightarrow m = \pm i$$

Therefore, the general solution of (1) is

$$y = c_1 cosx + c_2 sinx \qquad \dots \qquad (3)$$

where, c_1 and c_2 are arbitrary constants.

Using the boundary donation

$$y(0) = 1,$$

we get

$$1 = c_1 + 0$$

$$\therefore c_1 = 1$$

Again, using

$$y(\pi) = a,$$

we get

$$a = c_1 cos\pi + c_2 sin\pi = -c_1$$

$$a = -1$$

That is, second boundary condition requires that $a = -1$ for the BVP to have solution. If $a \neq -1$, the BVP has no solution.

On the other hand if $a = -1$, the BVP has infinitely many solution of the form $y = cosx + c_2 sinx$

for arbitrary c_2.

Example

Solve the boundary value problem

$$y" + 3y = 0, y(0) = 0, y(\pi) = 0$$

Solution

The given BVP is

$$y" + 3y = 0 \qquad \dots \qquad (1)$$

together with

$$y(0) = 0, y(\pi) = 0 \qquad \dots \qquad (2)$$

The A.E. of (1) is

$$m^2 + 3 = 0$$

$$\therefore\ m = \pm\sqrt{3}i = \sqrt{3}i, -\sqrt{3}i$$

Therefore, the general solution of (1) is

$$y = c_1 cos\sqrt{3}x + c_2 sin\sqrt{3}x \qquad \dots \qquad (3)$$

Using the condition

$$y(0) = 0,$$

we get

$$0 = c_1.1 + 0 \implies c_1 = 0$$

Again, using the second condition

$y(\pi) = 0,$

we get

$$0 = c_1 cos\sqrt{3}\pi + c_2 sin\sqrt{3}\pi$$

$$\implies c_2 sin\sqrt{3}\pi = 0, \quad \because c_1 = 0$$

$$\implies c_2 = 0, \quad \because sin\sqrt{3}\pi \neq 0$$

Therefore, the required solution of BVP is

$$y = 0$$

for cell x, which is the trivial solution.

Example

Solve the BVP $y" + y = 0, y(0) = 0, y(L) = 0$

Solution

The given BVP is

$$y" + y = 0 \qquad \dots \qquad (1)$$

together with

$$y(0) = 0, y(L) = 0 \qquad \dots \qquad (2)$$

The auxiliary equation of (1) is

$$m^2 + 1 = 0$$

$$\therefore m = i, -i$$

Therefore, the general solution of (1) is

$$y = c_1 cosx + c_2 sinx \qquad \dots \qquad (3)$$

Now, using the condition $y(0) = 0$, we get

$$0 = c_1.1 + 0$$

$$\therefore\ c_1 = 0$$

Next, using second condition $y(L) = 0$, we get

$$0 = c_1 cosL + c_2 sinL$$

$$\implies c_2 sinL = 0, \quad \because c_1 = 0$$

If $sinL \neq 0$, we must have $c_2 = 0$.
Therefore, the solution of BVP is

$$y = 0,$$

the trivial solution.
On the other hand, if $sinL = 0$, then, the solution of the BVP is

$$y = c_2 sinx$$

for arbitrary values of c_2.
Thus, in this case the BVP has infinitely many solutions.

Example

Solve the BVP $y" + 4y = cosx, y'(0) = 0, y'(\pi) = 0$

Solution

The given BVP is

$$y" + 4y = cosx \quad \dots \quad (1)$$

Together with

$$y'(0) = 0, y'(\pi) = 0 \quad \dots \quad (2)$$

The auxiliary equation of (1) is

$$m^2 + 4 = 0$$
$$\therefore m = 2i, -2i$$

Therefore,

$$C.F. = c_1 cos2x + c_2 sin2x$$

where, c_1 and c_2 are arbitrary constants.

To find P.I.

Let $Y(x) = Acosx + Bsinx$

$$\Longrightarrow Y'(x) = -Asinx + Bcosx$$
$$and\ Y''(x) = -Acosx - Bsinx$$

Putting these values in (1), we get

$$(-Acosx - Bsinx) + 4(-Asinx + Bcosx) = cosx$$
$$\Longrightarrow (-A + 4B)cosx + (B - 4A)sinx = cosx$$

Equating the coefficients of like terms, we get

$$-A + 4B = 1 \quad \dots \quad (i)$$

and

$$B - 4A = 0 \quad \dots \quad (ii)$$

Solving (i) and (ii), we get

$$A = \frac{1}{15}, B = \frac{4}{15}.$$
$$\therefore\ Y(x) = \frac{1}{15}cosx + \frac{4}{15}sinx$$

Hence the general solution of (1) is

$$y = C.F. + P.I.$$
$$y = c_1 cos2x + c_2 sin2x + \frac{1}{15}cosx + \frac{4}{15}sinx$$

Differentiating it, we get

$$y' = -2c_1 sin2x + 2c_2 cos2x - \frac{1}{15}sinx + \frac{4}{15}cosx$$

Using the condition $y'(0) = 0$, we get

$$0 = 0 + 2c_2 + 0 + \frac{4}{15}$$
$$\Rightarrow c_2 = -\frac{2}{15}$$
Next, using the condition
$$y'(\pi) = 0,$$
we get,
$$0 = 0 + 2c_2 + 0 - \frac{4}{15}$$
$$\Rightarrow c_2 = \frac{2}{15}$$
Hence, for arbitrary value of c_1
$$c_2 = -\frac{2}{15} \ or \ \frac{2}{15}$$
Therefore, the required solution of the BVP is
$$y = c_1 cos2x + \frac{2}{15} sin2x + \frac{1}{15} cosx + \frac{4}{15} sinx$$
Or,
$$y = c_1 cos2x - \frac{2}{15} sin2x + \frac{1}{15} cosx + \frac{4}{15} sinx.$$

Eigenvalue Problems

The matrix equation $Ax = \lambda x$ has the solution $x = 0$ for every value of λ. This solution is called the zero or trivial solution. But for some certain values of λ the equation has non-zero or non-trivial solutions, called the eigen vectors. The case is similar for BVP.

Consider the BVP
$$y'' + \lambda y = 0 \quad \ldots \quad (1)$$
together with
$$y(0) = 0, y(\pi) = 0 \quad \ldots \quad (2)$$
For $\lambda = 2$, equation (1) takes the form
$$y'' + 2y = 0$$
together with the condition (2) has only the trivial solution $y = 0$.

For $\lambda = 1$, equation (1) takes the form
$$y'' + y = 0$$
together with the conditions (2), has non-trivial solutions.

The values of λ, for which the BVP (1) and (2) have non-trivial solutions, are called the eigenvalues.

In above examples $\lambda = 1$ is an eigen-value and $\lambda = 2$ is not an eigenvalue.

The non-trivial solutions are called the eigenfunctions.

Eigenvalue Problems

Consider the boundary value problem
$$y'' + \lambda y = 0 \quad \ldots \quad (1)$$
together with
$$y(0) = 0, y(\pi) = 0 \quad \ldots \quad (2)$$
The A.E. of (1) is
$$m^2 + \lambda = 0$$
$$\Rightarrow m^2 = -\lambda \quad \ldots \quad (2)$$
We deal with the following three different cases.

Case I [when $\lambda = 0$]

If $\lambda = 0, m^2 = 0 \Rightarrow m = 0, 0.$

Therefore, the general solution of (1) is

$$y = e^{0x}(c_1 + c_2x) = c_1 + c_2x$$

$$\therefore y = c_1 + c_2x \quad \ldots \quad (4)$$

Using $y(0) = 0$, (4) gives $c_1 = 0$.

Again using $y(\pi) = 0$, (4) implies

$$0 = c_1 + c_2\pi$$

$$\Rightarrow 0 = 0 + c_2\pi \Rightarrow c_2 = 0.$$

Therefore, the required solution of the BVP is

$$y = 0, the\ trivial\ solution.$$

Case II [when $\lambda < 0 \Rightarrow -\lambda > 0$]

If $\lambda < 0, then - \lambda > 0.$

$$\therefore m^2 = -\lambda \Rightarrow m = \pm\sqrt{-\lambda}$$

So, the general solution of BVP is

$$y = c_1e^{\sqrt{-\lambda}x} + c_2e^{-\sqrt{-\lambda}x} \quad \ldots \quad (5)$$

Using $y(0) = 0$, (5) implies

$$c_1 + c_2 = 0 \quad \ldots \quad (6)$$

and using $y(\pi) = 0$, (5) implies

$$0 = c_1e^{\sqrt{-\lambda}\pi} + c_2e^{-\sqrt{-\lambda}\pi} \quad \ldots \quad (7)$$

Solving (6) and (7), we find

$$c_1 = 0, c_2 = 0$$

Hence, the solution of BVP is

$$y = 0,$$

which is also a trivial solution.

Case III [When λ > 0]

If λ > 0, then -λ < 0

$$\therefore m^2 = \lambda$$

$$\Rightarrow m = 0 \pm i\sqrt{\lambda}$$

Therefore, the general solution of (1) is

$$y = e^{0x}\left(c_1cos\sqrt{\lambda}x + c_2sin\sqrt{\lambda}x\right)$$

$$\Rightarrow y = c_1cos\sqrt{\lambda}x + c_2sin\sqrt{\lambda}x \quad \ldots \quad (8)$$

Using $y(0) = 0$, (8) implies

$$c_1 = 0.$$

∴ (8) becomes

$$y = c_2sin\sqrt{\lambda}x \quad \ldots \quad (9)$$

Using $y(\pi) = 0$, (9) reduces to

$$c_2sin\sqrt{\lambda}\pi = 0.$$

For non-trivial solution $c_2 \neq 0$.

Let us choose $c_2 = 0$, for particular.

$$c_2sin\sqrt{\lambda}\pi = 0$$

$$\Rightarrow sin\sqrt{\lambda}\pi = 0$$

$$\Rightarrow \sqrt{\lambda}\pi = n\pi$$

$$\Rightarrow \sqrt{\lambda} = n$$

$$\Rightarrow \lambda = n^2, for\ n = 1, 2, 3, \ldots$$

$\therefore \lambda_1 = 1^2, \lambda_2 = 2^2, \lambda_3 = 3^2, \quad \ldots \quad \lambda_n = n^2$ are eigenvalues of the differential equation (1) and (2) and the corresponding non-trivial solutions are

$y_1(x) = sinx, y_2(x) = sin2x, y_3(x) = sin3x, \quad \ldots \quad , y_n(x) = sinnx,$ which are the eigenfunctions of the BVP.

Example

Find the eigenvalues and eigenfunctions of the BVP

$$y'' - \lambda y = 0, y(0) = 0, y'(L) = 0.$$

Solution

The given BVP is

$$y'' - \lambda y = 0 \quad \ldots \quad (1)$$

together with

$$y(0) = 0, y'(L) = 0 \quad \ldots \quad (2)$$

The A.E. of (1) is

$$m^2 - \lambda = 0$$

$$\Rightarrow m^2 = \lambda$$

Case I [when λ = 0]

If λ = 0, we have

$$m^2 = 0$$

$$\therefore m = 0, 0$$

Therefore, the general solution of (1) is

$$y = c_1 + c_2 x \quad \ldots \quad (3)$$

Using $y(0) = 0$, we get $c_1 = 0$.

And $y' = c_2$, differentiating (3).

$\therefore$ $using\ y'(L) = 0, we\ get\ c_2 = 0.$

Therefore, the solution of the BVP is

$$y = 0, the\ trivial\ solution.$$

Case II [λ > 0]

If λ > 0, we have

$$m^2 = \lambda$$

$$\therefore m = \pm \sqrt{\lambda}.$$

The general solution of (1) is

$$y = c_1 e^{\sqrt{\lambda}x} + c_2 e^{-\sqrt{\lambda}x} \quad \ldots \quad (4)$$

Using $y(0) = 0$, (4) implies

$$0 = c_1 + c_2 \Rightarrow c_1 + c_2 = 0 \quad \ldots \quad (5)$$

Differentiating (4), we have

$$y' = \sqrt{\lambda} c_1 e^{\sqrt{\lambda}x} - \sqrt{\lambda} c_2 e^{-\sqrt{\lambda}x}$$

using $y'(L) = 0$, this implies

$$0 = \sqrt{\lambda}\left[c_1 e^{\sqrt{\lambda}L} - c_2 e^{-\sqrt{\lambda}L}\right]$$

$$c_1 e^{\sqrt{\lambda}L} - c_2 e^{-\sqrt{\lambda}L} = 0 \quad \ldots \quad (6)$$

Solving (5) and (6)

we get $c_1 = 0, c_2 = 0$

therefore, the required solution of the given BVP is

$$y = 0$$

again, a trivial solution

case III [λ < 0]

If λ < 0, then -λ > 0

$$\therefore\ m^2 = -\lambda$$

$$\Rightarrow m = 0 \pm i\sqrt{\lambda}$$

Therefore, the general solution of (1) is

$$y = c_1 cos\sqrt{\lambda}x + c_2 sin\sqrt{\lambda}x \quad \dots \quad (7)$$

$$\Rightarrow y' = -\sqrt{\lambda}c_1 s\sqrt{\lambda}xin + \sqrt{\lambda}c_2 cos\sqrt{\lambda}x \quad \dots \quad (8)$$

Using $y\ (0) = 0$, (7) implies

$$0 = c_1 + 0$$

$$\therefore c_1 = 0$$

Using this value in (8), we get

$$y' = \sqrt{\lambda}c_2 cos\sqrt{\lambda}x$$

Using $y'(L) = 0$, we get

$$0 = \sqrt{\lambda}c_2 cos\sqrt{\lambda}L$$

$$\Rightarrow c_2 cos\sqrt{\lambda}L = 0$$

For non-trivial solution $c_2 \neq 0$. And choose $c_2 = 1$ to get

$$cos\sqrt{\lambda}L = 0$$

$$\Rightarrow \sqrt{\lambda}L = (2n-1)\frac{\pi}{2}$$

$$\Rightarrow \sqrt{\lambda} = \frac{(2n-1)\pi}{2L}$$

$$\Rightarrow \lambda = \left[\frac{(2n-1)\pi}{2L}\right]^2$$

∴ the eigen values are given by

$$\lambda_n = \left[\frac{(2n-1)\pi}{2L}\right]^2$$

and the corresponding eigenfunctions are

$$y_n(x) = sin\left[\frac{(2n-1)nx}{2L}\right]^2$$

Fourier Series

Periodicity of sine and cosine Functions

Periodic Functions

A function f(x) is said to be periodic of period $\lambda > 0$ if the domain of f contains $x + \lambda$ whenever it contains x and if

$$f(x+\lambda) = f(x)$$

for every value of x. The smallest value of $\lambda > 0$ is called the fundamental period or simply the period of $f(x)$.

For example, $sinx$ has $2\pi, 4\pi, 6\pi,\ \dots$ as period and the first value 2π is the period of $sinx$.

Similarly, $cosx$ is also a periodic function of period 2π.

But $tanx$ is a periodic function of period π since $tan(x+\pi) = tanx$.

Example $sinnx$ is of period $\frac{2\pi}{n}$.

Properties of Periodic Functions

Let f and g be any two functions with a common period k. Then,

$$(i) \qquad f(x+nk) = f(x)\ \forall\, n \in Z^{+}$$

$$(ii) \quad c_1 f + c_2 g \; is\ periodic\ with\ period\ k$$

Let $F(x) = c_1 f(x) + c_2 g(x)$. Then,

$$F(x+k) = c_1 f(x+k) + c_2 g(x+k)$$
$$= c_1 f(x) + c_2 g(x) = F(x)$$

$$(iii) \qquad fg\ is\ perioid\ of\ k.$$

Let $F(x) = f(x)g(x)$. Then,

$$F(x+k) = f(x+k)g(x+k)$$
$$= f(x)g(x) = F(x)$$

(iv) The sum of any finite number of periodic functions with common period k is a periodic function of period k.

(v) The sum of the convergent infinite series with common period k is also a periodic function of periodic k.

(vi) $f(\alpha x)$ is a periodic function with period $\frac{k}{|\alpha|}$, where $\alpha \neq 0$ is any real number. For $sin\alpha x$ is a periodic function of period $\frac{2\pi}{|\alpha|}$.

Example

Determine whether the given function $sin\frac{\pi x}{L}$ is periodic or not. If periodic, find it's fundamental period.

Solution

Given: $f(x) = sin\frac{\pi x}{L}$

Let k be the fundamental period of $f(x)$. Then,

$$f(x+k) = f(x)$$
$$\Rightarrow sin\frac{\pi(x+k)}{L} = sin\frac{\pi x}{L}$$
$$\Rightarrow sin\left(\frac{\pi x}{L} + \frac{k\pi}{L}\right) = sin\frac{\pi x}{L}$$
$$\Rightarrow sin\left(\frac{\pi x}{L} + \frac{k\pi}{L}\right) = sin\left(\frac{\pi x}{L} + 2\pi\right)$$
$$\Rightarrow \frac{\pi x}{L} + \frac{k\pi}{L} = \frac{\pi x}{L} + 2\pi$$
$$\Rightarrow \frac{k\pi}{L} = 2\pi$$
$$\Rightarrow k = 2L$$

$\therefore\ sin\frac{\pi x}{L}$ is periodic with periodic $2L$.

Orthogonality of sine and cosine Functions

Inner Product

Let u and v be two real valued functions defined in an interval $a \leq x \leq b$. Then, the standard inner product of u and v, denoted by (u, v), is defined by

$$(u, v) = \int_a^b u(x)\, v(x) dx$$

Orthogonality

The functions u and v are said to be orthogonal on $[a, b]$ if

$$(u,v)=\int_a^b u(x)\,v(x)dx=0.$$

A set of functions $f_1(x), f_2(x),$ ... defined in an interval $[a,b]$ is said to be orthogonal set of functions if

$$\int_a^b f_m(x)\,f_n(x)dx=0, m\neq n.$$

This set is said to be orthonormal if

$$\int_a^b f_m(x)\,f_n(x)dx=\begin{cases}0, & if\;\; m\neq n\\ 1, & if\;\; m=n\end{cases}.$$

For example, the set

$$\left\{\frac{cosx}{\sqrt{\pi}},\frac{cos2x}{\sqrt{\pi}},\frac{cos3x}{\sqrt{\pi}},\qquad \dots\right\}$$

is an orthonormal if in $[-\pi,\pi]$.

Example

Show that the functions $sin\frac{m\pi x}{L}$ and $cos\frac{m\pi x}{L}$, $m\ = 1, 2, \dots$ form mutually orthogonal set of functions in an internal $[-L,L]$ with the following orthogonality relations.

$$a.\qquad \int_{-L}^{L} cos\frac{m\pi x}{L}cos\frac{n\pi x}{L}dx=\begin{cases}0, & m\neq n\\ L, & m=n\end{cases}$$

$$b.\qquad \int_{-L}^{L} cos\frac{m\pi x}{L}sin\frac{n\pi x}{L}dx=0\quad for\; all\; m,n$$

$$c.\qquad \int_{-L}^{L} sin\frac{m\pi x}{L}sin\frac{n\pi x}{L}dx=\begin{cases}0, & m\neq n\\ L, & m=n\end{cases}$$

Solution

Case I [when $m=n$]

a)

$$\int_{-L}^{L} cos\frac{m\pi x}{L}cos\frac{n\pi x}{L}dx$$

$$=\int_{-L}^{L} cos^2\left(\frac{m\pi x}{L}\right)dx$$

$$=\int_{-L}^{L}\left(\frac{1+cos\frac{2m\pi x}{L}}{2}\right)dx$$

$$=\frac{1}{2}\left[x+\frac{sin\frac{2m\pi x}{L}}{\frac{2m\pi}{L}}\right]_{-L}^{L}$$

$$=\frac{1}{2}\left[L+\frac{L}{2m\pi}sin2m\pi+L+\frac{L}{2m\pi}sin2m\pi\right]$$

$$= \frac{1}{2}[2L + 0 + 0] = L.$$

Therefore,

$$\int_{-L}^{L} cos\frac{m\pi x}{L} cos\frac{n\pi x}{L} dx = 0 \; if \; m = n.$$

Case II [when $m \neq n$]

$$\int_{-L}^{L} cos\frac{m\pi x}{L} cos\frac{n\pi x}{L} dx$$

$$= \frac{1}{2}\int_{-L}^{L} 2cos\frac{m\pi x}{L} cos\frac{n\pi x}{L} dx$$

$$= \frac{1}{2}\left[\int_{-L}^{L} cos\left(\frac{m\pi x}{L} + \frac{n\pi x}{L}\right) + cos\left(\frac{m\pi x}{L} - \frac{n\pi x}{L}\right)\right] dx$$

$$= \frac{1}{2}\frac{L}{\pi}\left[\frac{sin(m+n)\frac{\pi x}{L}}{m+n} + \frac{sin(m-n)\frac{\pi x}{L}}{m-n}\right]_{-L}^{L}$$

$$= \frac{1}{2}\frac{L}{\pi} \times 0 = 0$$

Hence,

$$\int_{-L}^{L} cos\frac{m\pi x}{L} cos\frac{n\pi x}{L} dx = 0, \qquad if \; m \neq n.$$

Similarly,

$$b. \qquad \int_{-L}^{L} cos\frac{m\pi x}{L} sin\frac{n\pi x}{L} dx = 0 \;\; for \; all \; m, n$$

$$c. \qquad \int_{-L}^{L} sin\frac{m\pi x}{L} sin\frac{n\pi x}{L} dx = \begin{cases} 0, & m \neq n \\ L, & m = n \end{cases}$$

The Euler-Fourier Formula

Trigonometric Series

A series of the form

$$\frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n cosnx + b_n sinnx)$$

is known as a trigonometric series where a_0, a_n, b_n are any real numbers.

Note

i. If $a_n = 0$, the series is pure sine series.

ii. If $b_n = 0$, the series is pure cosine series.

iii. $sin\alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \quad ... \quad + \sin(\alpha + (n-1)\beta) = \frac{sin\frac{n}{2}\beta}{sin\frac{\beta}{2}} sin\left\{\alpha + \left(\frac{n-1}{2}\right)\beta\right\}$

iv. $\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \quad \dots \quad + \cos(\alpha+(n-1)\beta) = \frac{\sin\frac{n}{2}\beta}{\sin\frac{\beta}{2}} \cos\left\{\alpha + \left(\frac{n-1}{2}\right)\beta\right\}$.

The Fourier Series

If the trigonometric series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right) \quad \dots \quad (1)$$

Converges to a function $f(x)$ whose value at each point x is the sum of the series for that value of x, then the seires (1) is said to be the fourier series of $f(x)$. The coefficients a_n, b_n are called the fourier coefficients.

Determination of the Fourier Coefficients

Consider the Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{m=1}^{\infty}\left(a_m \cos\frac{m\pi x}{L} + b_m \sin\frac{m\pi x}{L}\right) \quad \dots \quad (1)$$

where (1) is convergent to $f(x)$ in the interval $[-L, L]$.

From (1), taking definite integration from $-L$ to L, we get

$$\int_{-L}^{L} f(x)\,dx = \int_{-L}^{L} \frac{a_0}{2} dx + \sum_{m=1}^{\infty}\left(a_n \int_{-L}^{L} \cos\frac{m\pi x}{L} dx + b_n \int_{-L}^{L} \sin\frac{m\pi x}{L} dx\right)$$

$$= \frac{a_0}{2}[L-(-L)] + 0 + 0 = a_0 L$$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx.$$

Next, multiplying, both sides of (1) by $\cos\frac{n\pi x}{L}$ and integrating over $[-L, L]$, we get

$$\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L} dx = \frac{a_0}{2}\int_{-L}^{L} \cos\frac{n\pi x}{L} dx + \sum_{n=1}^{\infty}\left(a_m \int_{-L}^{L} \cos\left(\frac{m\pi x}{L}\right)\cos\left(\frac{n\pi x}{L}\right) dx + b_m \int_{-L}^{L} \sin\frac{m\pi x}{L}\cos\frac{n\pi x}{L} dx\right)$$

$$= 0 + a_m L + 0 = a_m L$$

$$a_m = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L} dx, \qquad n = 1,2,3, \quad \dots$$

And similarly,

$$b_m = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L} dx, \qquad n = 1,2,3, \quad \dots$$

Fourier Convergence Theorem

Theorem (Without Proof)

Let $f(x)$ and it's derivative $f'(x)$ be piecewise continuous in the interval $[-L, L]$. Moverover, let $f(x)$ be defined outside the interval $[-L, L]$ so that it is periodic with period $2L$. Then, $f(x)$ has a Fourier series:

$$f(x) = \frac{a_0}{2} + \sum_{m=1}^{\infty}\left(a_m \cos\frac{m\pi x}{L} + b_m \sin\frac{m\pi x}{L}\right).$$

The Fourier series converges to $f(x)$ at all points where $f(x)$ is continuous and to the mean

$$\frac{f(x+) + f(x-)}{2},$$

of right-hand and left-hand limits, at all points where $f(x)$ is discontinuous.

Even and Odd Functions

Even Function

A function $f(x)$ is an even function if it's domain contains $-x$ whenever it contains x and if $f(-x) = f(x)$.

Examples

$cosx, x^2, x^4 - 2x^2 + 1, |x|, x^{2n}$ are even functions.

Odd Function

A function $f(x)$ is said to be an odd function if its domain contains $-x$ whenever it contains x and if $f(-x) = -f(x)$.

Examples

$sinx, x^3, x^3 + 2x, x^{2n+1}$are odd functions.

Elementary Properties of Even and Odd Functions.

1. The sum (or difference) and product (or quotient) of two even functions is an even function.
2. The sum (or difference) of two odd functions is an odd function.
3. The product (or quotient) of two odd functions is an even function.
4. The sum (or difference) of an even function and an odd function is neither even nor odd.
5. The product (or quotient) of an even function and an odd function is an odd function
6. If f is an even function, then

$$\int_{-L}^{L} f(x)\,dx = 2\int_{0}^{L} f(x)\,dx$$

7. If $f(x)$ is an odd function, then

$$\int_{-L}^{L} f(x)\,dx = 0.$$